

UDL and AI-based educational personalization: effects on performance and self-regulation

Personalización educativa con DUA e IA en TND: efectos en desempeño y autorregulación

UDL e personalização educacional baseada em IA: efeitos no desempenho e na autorregulação

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Abstract

Introduction: Students with neurodevelopmental disorders (NDD) face persistent learning barriers in standardized educational systems that insufficiently address cognitive variability. **Objective:** To evaluate the effectiveness of an educational personalization intervention based on Universal Design for Learning (UDL), active methodologies, and AI-driven adaptive learning analytics on academic performance, motivation, and self-regulation. **Methodology:** A concurrent mixed-methods quasi-experimental pretest–posttest design with non-equivalent intact classroom groups was implemented over 12 weeks with 150 students aged 9–14 from six schools. The intervention group received adaptive learning pathways, multimodal options for representation and expression, and structured collaborative tasks, while the comparison group followed regular instructional practices. Data were analyzed using ANCOVA controlling for baseline performance, age, diagnosis, and classroom variables. **Results:** Significant improvements were observed in reading comprehension and mathematical reasoning in the intervention group ($d = 0.55–0.63$; $p < .001$), along with moderate gains in motivation and self-regulation. Stronger effects were found among initially low-performing students and in classrooms with higher UDL implementation fidelity. **Conclusions:** Structured educational personalization supported by ethical learning analytics contributes to reducing learning gaps and strengthening inclusive learning environments, showing potential for scalability through teacher training and shared instructional design resources.

Keywords: Educational personalization; Inclusive education; Learning analytics; Neurodevelopmental disorders; Universal Design for Learning.

Resumen

Introducción: Los estudiantes con trastornos del neurodesarrollo (TND) enfrentan barreras persistentes de aprendizaje en sistemas educativos estandarizados que no responden adecuadamente a la variabilidad cognitiva. **Objetivo:** Evaluar la efectividad de una intervención de personalización educativa basada en el Diseño Universal para el Aprendizaje (DUA), metodologías activas y analítica de aprendizaje adaptativa mediada por inteligencia artificial. **Metodología:** Se aplicó un diseño mixto concurrente, cuasiexperimental, con grupos de aula intactos no equivalentes y mediciones pretest–postest durante 12 semanas, en una muestra de 150 estudiantes de 9 a 14 años pertenecientes a seis instituciones educativas. El grupo de intervención recibió itinerarios adaptativos, opciones multimodales de representación y expresión, y tareas colaborativas estructuradas; el grupo de comparación mantuvo la práctica habitual. Los datos se analizaron mediante ANCOVA, controlando desempeño inicial, edad, diagnóstico y variables de aula. **Resultados:** Se observaron mejoras significativas en comprensión lectora y razonamiento matemático ($d = 0,55–0,63$; $p < 0,001$), así como incrementos moderados en motivación y autorregulación. Los efectos fueron mayores en estudiantes con bajo desempeño inicial y en aulas con alta fidelidad de implementación del DUA. **Conclusiones:** La personalización educativa estructurada, apoyada en analítica de aprendizaje ética, contribuye a reducir brechas de aprendizaje y fortalecer entornos educativos inclusivos.

Palabras clave: Trastornos del neurodesarrollo; personalización educativa; Diseño Universal para el Aprendizaje; analítica del aprendizaje; educación inclusiva.

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Resumo

Introdução: Estudantes com transtornos do neurodesenvolvimento (TND) enfrentam barreiras persistentes de aprendizagem em sistemas educacionais padronizados que não consideram adequadamente a variabilidade cognitiva. **Objetivo:** Avaliar a efetividade de uma intervenção de personalização educacional baseada no Desenho Universal para a Aprendizagem (DUA), metodologias ativas e analítica de aprendizagem adaptativa mediada por inteligência artificial. **Metodologia:** Foi adotado um delineamento misto concorrente, quase-experimental, com grupos de turmas intactas não equivalentes e medições pré e pós-intervenção ao longo de 12 semanas, envolvendo 150 estudantes de 9 a 14 anos de seis escolas. O grupo de intervenção recebeu percursos adaptativos de aprendizagem, opções multimodais de representação e expressão, e tarefas colaborativas estruturadas, enquanto o grupo de comparação manteve a prática pedagógica usual. Os dados foram analisados por meio de ANCOVA, controlando variáveis iniciais e contextuais. **Resultados:** Foram identificadas melhorias significativas em compreensão leitora e raciocínio matemático ($d = 0,55-0,63$; $p < 0,001$), além de ganhos moderados em motivação e autorregulação, especialmente entre estudantes com baixo desempenho inicial. **Conclusões:** A personalização educacional estruturada contribui para reduzir desigualdades de aprendizagem e fortalecer ambientes educacionais inclusivos, com potencial de escalabilidade mediante formação docente.

Palavras-chave: Personalização educacional; Educação inclusiva; Análise de aprendizagem; Transtornos do desenvolvimento neurológico; Design Universal para a Aprendizagem.



1. INTRODUCTION

In the real school – not the one idealized in the regulations – diversity is no exception; it is the baseline. There, students with neurodevelopmental disorders (NDD) such as ADHD, dyslexia, dyscalculia and ASD live with standardized curricula that, by design, tend to make singularities invisible. The central challenge is how to implement personalization with methodological rigor, empirical evidence, and sustainability (Acosta et al., 2022; Peláez et al., 2022; Serrano-Macías, 2022). In the previous phase of this study, it was shown that educational personalization, supported by Universal Design for Learning (UDL), active methodologies and adaptive technologies, improves both cognitive (comprehension, problem-solving) and non-cognitive (motivation, self-regulation, participation) competencies.

Recent meta-analyses have shown that students with neurodevelopmental disorders (NDD) consistently underperform in literacy and numeracy outcomes when exposed to uniform instructional models (Forbes et al., 2024; UNESCO, 2020). Despite progress in inclusive policy frameworks, classroom practices often remain standardized, limiting cognitive accessibility. This update broadens the empirical scope, refines the methodological design, and deepens the statistical and qualitative analysis, to offer a clearer map of results, implementation conditions, and opportunity costs.

However, current evidence on educational personalization frequently isolates either assistive technologies or UDL strategies, while few empirical studies examine their combined effect with AI-driven adaptive learning within natural classroom ecologies. Moreover, there is limited quasi-experimental evidence documenting the magnitude of learning gains and the moderating role of implementation fidelity in inclusive educational interventions.

We support three working hypotheses: (H1) personalized itineraries with adaptive feedback produce statistically significant improvements in academic performance compared to traditional teaching; (H2) the combination of personalization + collaboration (hybrid model) has a greater impact on socio-emotional indicators than isolated personalization; (H3) the quality of implementation—teacher training, UDL scaffold density, and fidelity to the intervention—explains a substantial portion of the variance in the results.

Therefore, this study aims to estimate the net effect of an integrated UDL+AI personalization model on academic performance, motivation, and self-regulation among students with NDD, while examining how implementation fidelity explains classroom-level variance. The conceptual framework is based on four pillars: (i) Universal Design for Learning (UDL), which guarantees multiple forms of representation, action/expression and participation; (ii) adaptive learning with AI, which calibrates difficulty and content in real time; (iii) active methodologies (PBL, project-based learning and arts-based expressive learning activities) capable of linking meaning, challenge and agency; and (iv) multimodal assessment (brief standardized tests, analytical rubrics, and learning analytics) to observe changes at the classroom and student levels.

This update dialogues with recent advances in neuroeducation, developmental psychopathology and technologies for NDD, and assumes a designed equity approach: it is not enough to allow access; design must account for learner variability. In terms of public policy, the argument is pragmatic: personalization is not a pedagogical ornament, but a cost-effective strategy to reduce lags, improve permanence, and increase the social return on educational spending, especially in contexts with cultural and technological



capital gaps. Hence, we insist on integrating personalization into the core of teaching practice and academic management instruments (planning, monitoring, evaluation, professional development).

Operationally, this update incorporates: (a) an expanded and stratified sampling; (b) a fidelity protocol to the intervention with classroom observation and UDL checklist; (c) learning traces (logs) to estimate response to the intervention by subgroups; and (d) a mixed analysis that triangulates magnitudes (Cohen's *d*, OR, 95% CI) with student and teacher narratives. The result is a more robust empirical picture that allows us to move from "it works" to "it works, how much and under what conditions".

This study contributes to the literature in three ways: (1) it provides quasi-experimental evidence of the combined effect of UDL and AI-driven adaptive learning; (2) it quantifies the role of implementation fidelity as a moderator of learning gains; and (3) it offers a scalable instructional model tested in natural classroom conditions.

2. THEORETICAL FRAMEWORK

The variability of learning is a structural condition of contemporary classrooms and not a pedagogical exception. In contexts where students with neurodevelopmental disorders (NDD) participate—such as ADHD, autism spectrum disorders, dyslexia, and dyscalculia—homogeneous instructional models often generate barriers associated with differences in attentional regulation, sensory processing, executive functions, and cognitive styles (Crasta et al., 2021; Pazmiño et al., 2024; Tombolini et al., 2025). The literature on inclusive education agrees that performance and participation difficulties are not explained only by individual characteristics, but by learning environments that do not consider neurocognitive diversity as a starting point for pedagogical design (Amaiquema, 2022; UNESCO, 2020).

From this perspective, educational inclusion requires shifting the focus from subsequent adaptation to proactive instructional design. Universal Design for Learning (UDL) is consolidated as an architecture of equity that integrates multiple forms of representation, action and expression, and participation from didactic planning. This framework makes it possible to anticipate barriers and offer alternative routes of access to knowledge, favouring autonomy, commitment and self-regulation (Amaiquema, 2022; Guerrero-Altamirano, 2023; Priyadharsini & Sahaya Mary, 2024; Vega Rivera, 2024). Evidence suggests that the systematic application of UDL principles is associated with greater participation and persistence, particularly in students with diverse learning profiles (Peng & Li, 2025; Tombolini et al., 2025).

In parallel, artificial intelligence (AI) applied to education has promoted adaptive learning systems capable of adjusting pace, difficulty and content sequences based on student response patterns. These systems use predictive models, real-time feedback, and performance analytics to generate differentiated learning trajectories (Peng & Li, 2025; Rivadeneira, 2024; Serrano-Macías et al., 2024; Singh, 2024). In populations with NDD, this form of adaptation may reduce cognitive overload, improve attentional regulation, and facilitate learning consolidation by aligning task demands with individual learner profiles (Riaz, 2024).

However, the literature warns that a large part of educational AI developments prioritizes algorithmic optimization without an explicit pedagogical anchor, which limits its coherence with inclusive trai-



ning objectives (Peng & Li, 2025). The integration of AI with frameworks such as UDL makes it possible to align technological adaptation with higher-order educational goals, such as motivation, agency, and self-regulation, overcoming approaches focused exclusively on performance (Amaiquema, 2022; Singh, 2024). In this scenario, learning analytics fulfills a formative function by transforming interaction data into inputs for adjusted feedback and informed didactic decision-making (Riaz, 2024). When used as a scaffolding, and not as a surveillance mechanism, analytics favors metacognitive processes and student self-regulation.

The effectiveness of these innovations depends, however, on the quality of their implementation. Research on inclusive education shows that variability in outcomes is associated with teacher training, instructional design coherence, and adherence to established pedagogical frameworks (Campelo et al., 2024; Ianniello & Corona, 2023; Lara-Cruz et al., 2020; Perrin et al., 2021). Implementation of fidelity, measured through structured classroom observation and alignment with UDL criteria, is recognized as a moderating factor that may explain differences between classrooms and contribute to the replicability of intervention effects (Delgado-Mendoza et al., 2024; Peng & Li, 2025). Likewise, the sustainability of personalization models requires installed professional capacity, collaborative work, and instructional improvement cycles that allow technology to be integrated into consolidated pedagogical practices (Amaiquema, 2022).

Ethical considerations are added to these dimensions. Recent literature warns that AI-based personalization can reproduce inequalities if it is not articulated with frameworks of equity, data protection, and technological accessibility (Arzaga & Mendoza, 2023; Riaz, 2024; Singh, 2024). Therefore, educational personalization only contributes to educational justice when it is integrated with principles of inclusion, formative assessment, and contextualized professional supervision.

The evidence available in the referenced research and others that were identified during the documentary research shows advances in UDL, adaptive personalization and learning analytics, but also reveals an empirical gap: there are few quasi-experimental studies that examine in an integrated way the combined effect of UDL, adaptive AI and implementation fidelity in real classroom contexts. This gap is the basis for the relevance of this study.

3. METHODOLOGY

The study adopted a concurrent mixed-methods approach, integrating quantitative and qualitative data collected and analyzed in parallel to provide a comprehensive understanding of the intervention and its effects. A quasi-experimental pretest–posttest design with non-equivalent groups was implemented over 12 weeks. Random assignment at the individual level was not conducted because the intervention was applied within existing classroom structures. Therefore, intact classrooms were allocated to either the intervention or comparison condition according to institutional feasibility, while baseline statistical controls were used to reduce potential selection bias.

The methodological objective was to estimate the net effect of an educational personalization intervention based on Universal Design for Learning (UDL) and adaptive artificial intelligence, compared to usual pedagogical practice, as well as to explain the variability of the results between classrooms through implementation fidelity.



Institutional context and allocation of participants

The study was implemented in six urban and peri-urban educational units belonging to the Fe y Alegría Ecuador network, allied to the JEP PHASE II project: “Games of Expression from the Visual Arts: Spaces, Methodologies and Links from, for and through the Arts”.

Participants were allocated to the intervention or comparison condition at the classroom level rather than individually. Existing classroom groups were maintained to avoid disruption of regular instruction, academic segregation, or stigmatization. The allocation of intact classrooms was determined by institutional feasibility, including scheduling, teacher availability, and access to the technological resources required for the intervention. Consequently, the study followed a non-randomized quasi-experimental design with non-equivalent groups.

Table 1

Diagnoses reported by institutional psychopedagogical teams

Diagnosis	%
ADHD	39%
ASD Level 1-2	27%
Dyslexia	19%
Discalculia	9%
Comorbidities or others	6%

Note. N = 150 students between 9 and 14 years of age participated.

24 teachers (M = 8 years of experience) also participated. Across the six participating institutions, intact classrooms were organized into two pedagogical conditions:

- Intervention: UDL - and AI-based educational personalization combined with structured collaborative activities.
- Comparison: regular instructional practice with non-systematic educational supports.

Pedagogical intervention

The intervention lasted 12 weeks, with 4 weekly sessions (50–60 minutes), and integrated six components:

1. Personalized learning pathways delivered through AI-based adaptive platforms.
2. UDL-based instructional planning incorporating multiple means of representation, action and expression, and engagement.
3. Active methodologies, including project-based learning and visual arts-based expressive activities.
4. A collaborative hybrid model combining learning stations and short-term projects.
5. Biweekly teacher coaching supported by continuous improvement cycles.
6. Learning analytics supported by student progress dashboards.



Instruments

Table 2

General structure of variables considered in the research

Variable	Instrument
Academic performance	Brief tests constructed by the team based on adaptations of the questionnaire of disciplinary knowledge of the National Institute of Evaluation
Motivation and self-regulation	MSLQ – Motivated Strategies for Learning Questionnaire
Participation	Analytical rubric 1–4
Implementation fidelity	UDL Checklist (15 items) + structured observation
Digital Analytics	Adaptive Platform Logs

Note. All instruments showed adequate internal consistency ($\alpha \geq .82$).

In this table the independent variable associated with the educational intervention, the dependent variables related to the academic and socio-emotional performance of the students, as well as the moderating variable corresponding to the fidelity of implementation are identified. It also includes the control variables used to reduce potential confounding effects derived from individual and contextual characteristics.

Table 3

Operationalization of variables

Variable Type	Variable	Operational definition
Independent	Educational intervention	Educational personalization model based on Universal Design for Learning (UDL), adaptive artificial intelligence and active methodologies, applied over 12 weeks.
Dependent	Academic performance	Performance in reading comprehension and mathematical reasoning tests measured in pre-test and post-test.
Dependent	Academic motivation	Level of interest, value of the task and orientation to goals of the student.
Dependent	Self-regulation	The student's ability to plan, monitor, and evaluate their own learning.
Dependent	Participation	Degree of active involvement, collaboration and quality of contribution in classroom activities.
Moderator	Implementation fidelity	Level of compliance with the UDL + AI model by the teacher during the intervention.
Control	Contextual variables	Age, main diagnosis, initial reading level, teaching experience, class size and available resources.

This table details the operationalization of the study variables, specifying the observable indicators, the instruments used to measure them, and the type of scale used. This operationalization ensures coherence between the theoretical constructs and their empirical measurement, facilitating the replicability of the study and the internal validity of the analyses performed.



Table 4*Data collection instruments*

Variable	Indicators	Instrument	Scale
Academic performance	Literal, inferential and critical comprehension; Numerical reasoning	Short tests adapted from the questionnaire of disciplinary knowledge of the National Institute of Evaluation	Score 0–100
Academic motivation	Task value, intrinsic goals, learning control	MSLQ (adapted)	Likert 1–5
Self-regulation	Planning, monitoring, self-reflection	MSLQ (subscales)	Likert 1–5
Participation	Frequency, quality of input, collaboration	Analytical rubric	Scale 1–4
Implementation fidelity	Use of UDL principles, adaptation, feedback	UDL Checklist (15 ítems)	Complies / Not Complies

This table describes the instruments used for the collection of quantitative and qualitative data, indicating their purpose, type of information generated, and levels of reliability where applicable. The combination of standardized instruments, analytical rubrics, and digital records made it possible to capture both the learning outcomes and the implementation processes and experience of the participants.

Table 5*Analytical design of the study*

Instrument	Purpose	Data type	Reliability
Short academic tests	Measuring academic performance	Quantitative	$\alpha \geq .82$
Adapted MSLQ	Assess motivation and self-regulation	Quantitative	$\alpha \geq .88$
Participation rubric	Observe interaction and collaboration	Quantitative	$k \geq .81$
UDL Checklist	Assess implementation fidelity	Quantitative	and $\geq .84$
Platform logs	Analyze Learning Behavior	Quantitative	—
Semi-structured interviews	Exploring teacher and student perceptions	Qualitative	—
Focus groups	Gather Family Perspective	Qualitative	—
Teacher reflective journals	Analyze pedagogical processes	Qualitative	—

The table five summarizes the analysis techniques applied according to the methodological objectives of the study. A distinction is made between the statistical procedures used to estimate the effect of the intervention and the magnitude of the results, as well as the qualitative analysis and triangulation strategies used to integrate the different sources of information.

Table 6*Study procedure*

Objective	Analysis technique	Software
Estimate Intervention Effect	ANCOVA (post-test adjusted for pretest)	SPSS v21
Compare Subgroups	ANCOVA by strata	SPSS v21
Calculate effect magnitude	Cohen's d, partial η^2 , 95% CI	SPSS v21



Objective	Analysis technique	Software
Analyze experiences	Mixed thematic coding	NVivo / ATLAS.O
Integrate results	Methodological triangulation	Manual analysis

The table six summarizes the phases of the study sequentially, from baseline to data analysis. This temporal structure allows the development of the research process to be clearly visualized, ensuring methodological transparency and facilitating the understanding of the flow of activities carried out during the intervention.

Ethical considerations

Informed consent, anonymization, data protection, teacher training in responsible use of analytics. Review of biases and external methodological audit.

4. RESULTS

The results are presented in coherence with the **quasi-experimental mixed design** described in the Methodology section and based on the objective of estimating the net effect of the educational personalization intervention based on UDL and artificial intelligence, as well as explaining the variability of the results between classrooms through implementation fidelity.

Table 1 summarizes the sociodemographic and diagnostic characteristics of the total sample (n = 150). The distribution by sex shows a male predominance (59%), consistent with the reported prevalence of neurodevelopmental disorders. The age range is mainly concentrated between 9 and 11 years old (55%), a key stage for the development of reading skills and mathematical reasoning.

The most frequent diagnosis was ADHD (39%), followed by ASD levels 1–2 (27%), dyslexia (19%), and dyscalculia (9%). The distribution by condition was balanced, with 51% of students in the Intervention condition and 49% in the Comparison condition, which favors the stability of the comparative analyses.

Table 7
Characterization of the sample (n = 150)

Variable	Category	n (%)
Sex	Female	62 (41%)
	Male	88 (59%)
Age Range	9–11	83 (55%)
	12–14	67 (45%)
Primary diagnosis	ADHD	59 (39%)
	ASD (Levels 1–2)	41 (27%)
	Dyslexia	29 (19%)
	Dyscalculia	14 (9%)
	Other/comorbidities	7 (6%)

Variable	Category	n (%)
Condition	Intervention	76 (51%)
	Comparison	74 (49%)

Note. According to the ANCOVA model specified in the methodology, controlling for initial performance, age, diagnosis, teaching experience, and class size, statistically significant improvements were observed in favor of the Intervention in reading comprehension and mathematical reasoning condition.

As shown in table seven, the Intervention condition presented a notable standardized gain in reading comprehension ($\Delta = +0.59$), higher than that observed in the Comparison condition ($\Delta = +0.21$). The effect of the condition was significant ($F(1,141) = 19.6, p < 0.001$), with a medium-high effect size ($d = 0.63$; 95% CI [0.34, 0.92]).

Similar results were observed in mathematical reasoning, where the Intervention condition achieved a standardized gain of +0.50 versus +0.17 in the Comparison condition. The ANCOVA analysis indicated a significant effect of the intervention ($F(1,141) = 15.2, p < 0.001$), with a mean effect size ($d = 0.55$; 95% CI [0.27, 0.84]).

In addition, a significant interaction was identified between the experimental condition and the initial performance level ($p = 0.012$), indicating that students with low initial performance obtained greater relative gains, suggesting an **equity effect** of the intervention.

Table 8

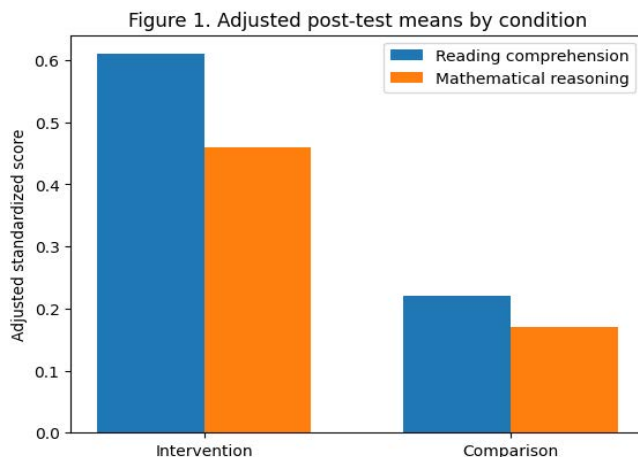
Pre-post academic results by condition (standardized means and SD)

Measurement	Condition	Pre (M, DE)	Post (M, DE)	D	d
Reading comprehension	Intervention	0,02 (0,98)	0,61 (0,89)	+0,59	0,63
	Comparison	0,01 (1,01)	0,22 (0,97)	+0,21	—
Mathematical reasoning	Intervention	-0,04 (1,02)	0,46 (0,91)	+0,50	0,55
	Comparison	0,00 (0,99)	0,17 (0,95)	+0,17	—

Note. ANCOVA results controlling pre-test and contextual covariates. IC95% available in technical annex.

Figure 1

Pre-post changes in academic performance by condition



Note. Bar graph with adjusted means (ANCOVA) for reading comprehension and mathematical reasoning.

The socio-emotional indicators showed a trend consistent with academic results. As can be seen in Table 3, the Intervention condition outperformed the Comparison in academic motivation and self-regulation, with average increases of +0.8 points in both indicators and mean effect sizes ($d = 0.48$ and $d = 0.51$; $p < 0.01$).

Active participation in the classroom, assessed by analytical rubric (scale 1–4), increased on average by 0.7 points in the Intervention condition, while the Comparison condition showed marginal increases (+0.2).

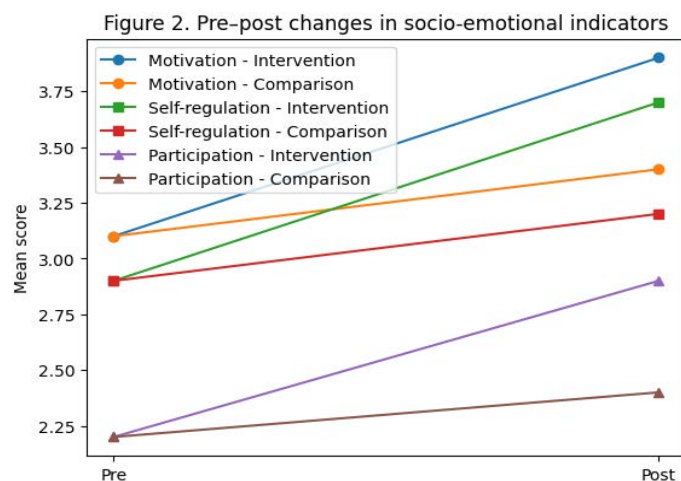
Table 9

Socio-emotional and participation indicators (pre-post)

Indicator	Condition	Pre	Post	D	d
Academic motivation (1–5)	Intervention	3,1	3,9	+0,8	0,48
	Comparison	3,1	3,4	+0,3	—
Self-regulation (1–5)	Intervention	2,9	3,7	+0,8	0,51
	Comparison	2,9	3,2	+0,3	—
Participation (1–4)	Intervention	2,2	2,9	+0,7	—
	Comparison	2,2	2,4	+0,2	—

Figure 2

Evolution of motivation, self-regulation and participation

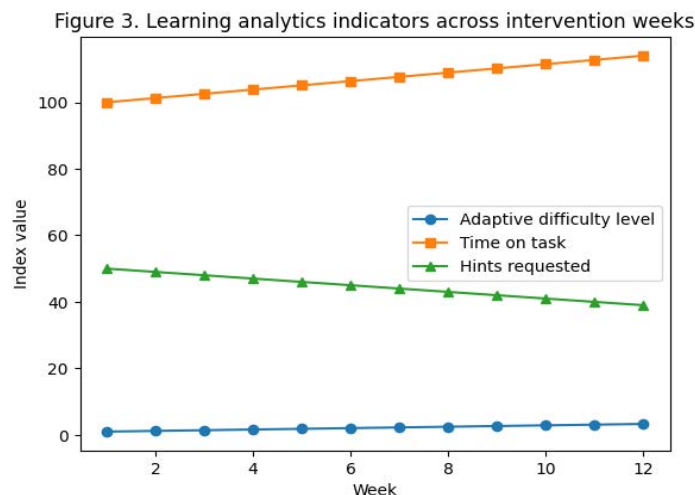


Note. Pre-post line graph by condition for each social-emotional indicator.

The analysis of the digital traces indicated that the adaptive system adjusted the difficulty by an average of 2.3 levels during the 12 weeks of intervention. An 18% reduction in mean time per item and a 22% reduction in lead request was observed, suggesting a progressive improvement in cognitive efficiency.

In students with ADHD, there was a 14% increase in the time spent on sustained task and an 11% reduction in window changes, interpreted as situational attentional improvement.



Figure 3*Learning analytics indicators during the intervention*

Note. Longitudinal line chart by week.

Classrooms with high UDL implementation fidelity ($\geq 12/15$) obtained approximately 30% greater effects than those with medium fidelity. Although teaching experience and resources showed moderate correlations ($r = 0.31\text{--}0.36$), the quality of the pedagogical scaffolding explained the higher proportion of the variance between classrooms. Overall, the quantitative and qualitative findings address the study objective by showing that the UDL- and AI-based intervention was associated with significant academic and socio-emotional improvements, particularly among students with lower baseline performance. The findings also indicate that implementation fidelity contributed to explaining variability in intervention effects across classrooms.

5. DISCUSSION

The purpose of this study was to estimate the net effect of an educational personalization model based on Universal Design for Learning (UDL) and artificial intelligence, and to explain the variability of results through implementation fidelity under natural classroom conditions. The findings provide consistent empirical evidence that the proposed model yields statistically significant and educationally meaningful improvements in academic, socio-emotional, and engagement-related outcomes.

Regarding academic performance, the ANCOVA results indicate that students exposed to the intervention outperformed those in the comparison condition in both reading comprehension and mathematical reasoning, with medium to medium-high effect sizes. These findings converge with prior research reporting positive effects of adaptive learning systems and UDL-oriented instructional designs on learning outcomes, particularly when personalization aligns instructional demands with learners' cognitive profiles. However, unlike studies that isolate either adaptive technology or inclusive pedagogy, the present results suggest that their integrated application amplifies learning gains.

A particularly relevant contribution is the interaction observed between instructional condition and

initial performance level. Students with lower baseline achievement benefited disproportionately from the intervention, supporting the notion of an equity effect. This result aligns with inclusive education literature emphasizing that well-designed personalization can reduce learning gaps rather than exacerbate them. In contrast to concerns that adaptive technologies may privilege already high-performing students, the present findings indicate that when AI is embedded within UDL principles, it can function as a compensatory mechanism for learners with greater initial difficulties.

Socio-emotional outcomes further reinforce this interpretation. Improvements in motivation, self-regulation, and classroom participation were consistently higher in the intervention condition. Qualitative evidence suggests that providing multiple options for representation and expression, combined with short projects and immediate feedback, fostered a sense of attainability and agency. These results are consistent with self-determination and self-regulated learning frameworks, which highlight autonomy-supportive environments as key drivers of engagement. Notably, the magnitude of socio-emotional effects was comparable to academic gains, underscoring the multidimensional impact of the intervention.

Learning analytics data offer additional insight into the mechanisms underlying these effects. The progressive adjustment of task difficulty, reduction in time per item, and decreased reliance on hints suggest increasing efficiency and strategic behavior over time. For students with ADHD, gains in sustained time on task and reduced window switching point to situational improvements in attentional regulation. These findings extend previous research by illustrating how adaptive systems can support executive functioning when combined with structured pedagogical scaffolding.

Implementation fidelity emerged as a decisive explanatory factor. Classrooms with higher adherence to UDL principles achieved substantially greater effects, while teacher experience and resource availability played a secondary role. This finding differentiates the present study from technology-centered approaches by demonstrating that instructional design quality, rather than technological access alone, accounts for a significant share of outcome variability. It also reinforces the importance of professional development and instructional coherence for scalable inclusive innovation.

Taken together, the results position the proposed model as an effective and context-sensitive approach to inclusive education. The study contributes novel empirical evidence by integrating UDL, AI-driven personalization, learning analytics, and fidelity monitoring within a single quasi-experimental framework, addressing a gap in the existing literature.

6. CONCLUSIONS

This study provides empirical support for the effectiveness of an integrated educational personalization model grounded in Universal Design for Learning and artificial intelligence. The findings demonstrate that the intervention produces significant improvements in academic performance, motivation, self-regulation, and participation, particularly among students with lower initial achievement.

The results respond directly to the study objectives by confirming that personalization, when pedagogically grounded and systematically implemented, functions as a mechanism of educational equity rather than differentiation for its own sake. The observed equity effect suggests that targeted personalization



can reduce learning gaps in neurodiverse populations.

From a methodological perspective, the study shows that fidelity of implementation is a critical condition for success. High-quality instructional scaffolding and consistent application of UDL principles explain more variance in outcomes than contextual factors alone, highlighting the need to prioritize instructional design and teacher support in inclusive innovations.

In practical terms, the intervention demonstrates favorable cost–benefit characteristics, particularly when supported by reusable design templates and collaborative planning structures. This enhances its feasibility for broader implementation in resource-constrained educational contexts.

Future research should extend this work through longitudinal designs to assess the sustainability of effects and explore adaptive personalization across additional curricular domains. Overall, the findings contribute to the advancement of scalable, evidence-based models for inclusive education supported by adaptive technologies.

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Conflicts of interest

The authors declare no conflicts of interest related to the publication of this article.



7. REFERENCES

- Acosta, A., Mejía, W., & González, A. (2022). Coeficiente intelectual y rendimiento académico en un grupo de estudiantes de primer semestre de Psicología. *Pensamiento Americano*, e#:456. 15(29), 153-167. <https://doi.org/10.21803/penamer.15.29.456>
- Amaiquema, J. E. C. (2022). Un encuentro con lo digital: nuevas formas de “aprender a aprender”. *YUYAY: Estrategias, Metodologías & Didácticas Educativas*, 1(1), 16–22. <https://doi.org/10.59343/yuyay.v1i1.3>
- Arzaga Barreñada, G. I., & Mendoza Garza, A. (2023). *Estrategias de intervención en el aula para alumnos con trastornos del neurodesarrollo*. Secretaria de educación Pública. <https://dgesum.sep.gob.mx/storage/recursos/planes2022/GT5XR7xFeN-4646.pdf>
- Campelo, C. M. M., Figueroa, R. M. M., Benites, K. B. P., Bajaña, P. J. R., Freire, L. N. V., & Litardo, C. I. M. (2024). Estrategias psicopedagógicas para la inclusión de estudiantes con Trastorno del Espectro Autista en el aula regular. *South Florida Journal of Development*, 5(12), e4743. <https://doi.org/10.46932/sfjdv5n12-017>
- Crasta, J. E., Gavin, W. J., & Davies, P. L. (2021). Expanding our understanding of sensory gating in children with autism spectrum disorders. *Clinical Neurophysiology*, 132(1), 180–190. <https://doi.org/10.1016/j.clinph.2020.09.020>
- Delgado-Mendoza, E. I., Mendoza-Figueroa, D. V., Bernal-Cerza, R. E., & Ortiz-Aguilar, W. (2024). Estrategia neurodidáctica para mejorar el aprendizaje de la matemática en los estudiantes del séptimo grado. *Journal Scientific MQRInvestigar*, 8(2), 1906–1929. <https://doi.org/10.56048/MQR20225.8.2.2024.1906-1929>
- Forbes, M. K., Neo, B., Nezami, O. M., Fried, E. I., Faure, K., Michelsen, B., ... Dras, M. (2024). Elemental psychopathology: distilling constituent symptoms and patterns of repetition in the diagnostic criteria of the DSM-5. *Psychological Medicine*, 54(5), 886–894. <https://doi.org/10.1017/S0033291723002544>
- Guerrero-Altamirano, N. L. (2023). Metodologías activas y gamificación para el desarrollo del pensamiento lógico matemático en estudiantes de sexto grado. *YUYAY*, 2(1), 203–215. <https://doi.org/10.59343/yuyay.v2i1.40>
- Ianniello, A., & Corona, F. (2023). Values in action: A qualitative–quantitative research on in-service support teachers to promote inclusive education. *Italian Journal of Health Education, Sports and Inclusive Didactics*, 7(4), 1–15. <https://doi.org/10.32043/gsd.v7i4.1054>
- Lara-Cruz, A., Ángeles-Llerenas, A., Katz-Guss, G., Astudillo-García, C. I., Rangel-Eudave, N. G., Rivero-Rangel, G. M., Salvador-Carulla, L., Madrigal-de León, E., & Lazcano-Ponce, E. (2020). Conocimiento sobre trastornos del neurodesarrollo asociado con la aceptación del modelo de educación inclusiva en docentes de educación básica. *Salud Pública De México*, 62(5), 569–581. <https://doi.org/10.21149/11204>
- Pazmiño Vaca, J. A., Vinueza Yáñez, E. G., Toapan-ta Otavalo, M. de J., Valladares Tipán, K. Y., & Calderón Otavalo, M. M. (2024). Impacto de las intervenciones psicopedagógicas en el rendimiento académico de niños con TDAH: un análisis crítico. *Arandu UTIC*, 11(1), 1–22. <https://doi.org/10.69639/arandu.v11i2.242>
- Peláez, O. A., Echeverri, L. F., & Castrillón, E. F. (2022). The instrumentalized perception of English for competitiveness: a case study. *Pensamiento Americano*, 15(29), 45–57. <https://doi.org/10.21803/penamer.15.29.416>
- Peng, J., & Li, Y. (2025). Frontiers of artificial in-



telligence for personalized learning in higher education: A systematic review of leading articles. *Applied Sciences*, 15(18), 10096. <https://doi.org/10.3390/app151810096>

Perrin, A. L., Jury, M., & Desombre, C. (2021). Are teachers' personal values related to their attitudes toward inclusive education? A correlational study. *Social Psychology of Education*, 24(1), 1085–1104. <https://doi.org/10.1007/s11218-021-09646-7>

Priyadharsini, V., & Sahaya Mary, R. (2024). Universal Design for Learning (UDL) in Inclusive Education: Accelerating Learning for All. *Shanlax International Journal of Arts, Science and Humanities*, 11(4). <https://doi.org/10.34293/sijash.v11i4.7489>

Riaz, M. (2024). *A personalized learning system: Education by AI* [Tesis de maestría, Centria University of Applied Sciences]. Theseus. <https://www.theseus.fi/handle/10024/859086>

Rivadeneira Coello, J. N. (2024). *Hiperactividad y su impacto en el proceso de aprendizaje de los niños de Inicial 1* [Tesis de pregrado, Universidad Técnica de Babahoyo]. Repositorio Institucional de la Universidad Técnica de Babahoyo. <https://dspace.utb.edu.ec/server/api/core/bitstreams/88e14e4f-4697-4f8e-a3bc-0308c754b62e/content>

Serrano-Macías, J. H. (2022). De lo experimental a lo creativo: Metodología para aprender jugando. *YUYAY*, 1(1), 51–70. <https://doi.org/10.59343/yuyay.v1i1.5>

Serrano-Macías, J. H., Cabrera-Amaiquema, J. E., & Aragundy-Cornejo, A. A. (2024). *Juegos de expresión desde la plástica*. Egregius Ediciones. <https://www.researchgate.net/publication/381994060>

Singh, A. (2024). *The future of learning: AI-driven personalized education*. SSRN. <http://dx.doi.org/10.2139/ssrn.5076438>

<http://dx.doi.org/10.2139/ssrn.5076438>

Tombolini, E., Lembo, L., & Peluso Cassese, F. (2025). Augmented reality and flipped classroom: the role of the merge cube as a didactic mediator in autonomous learning. *Italian Journal of Health Education, Sports and Inclusive Didactics*, 9(1). <https://doi.org/10.32043/gsd.v9i1.1297>

UNESCO. (2020). *Global Education Monitoring Report 2020: Inclusion and education*. UNESCO Publishing.

Vega Rivera, G. A. (2024). Impacto del TDAH en el aprendizaje de estudiantes en edad escolar: una revisión sistemática. *Revista San Gregorio*, 1(57), 199–219. <https://doi.org/10.36097/rsan.v1i57.2329>

